Physics Reference Notes

Ferdinand Tölkes

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Contents

1 Variational fermionic wave functions on lattices

1.1 Summary

1.2 Crash course on the origin of antisymmetric wave functions

See any introductory book on quantum mechanics, e.g. Schwabl, *Quantum Mechanics*, section 13.1.

The Heisenberg uncertainty relation implies that identical particles are indistinguishable. Consequently the probability amplitude $|\psi(x_1,\ldots,x_N)|^2$ of a quantum many-body state is invariant under the exchange $x_i \leftrightarrow x_j$ of two indistinguishable particles. Define the exchange operator

$$P_{ij}\,\psi(x_1,\ldots,x_i,\ldots,x_j,\ldots,x_N) \coloneqq \psi(x_1,\ldots,x_j,\ldots,x_i,\ldots,x_N), \tag{1}$$

so that invariance of the amplitude gives

$$P_{ij}\,\psi(\vec{x}) = \lambda\,\psi(\vec{x}), \qquad |\lambda|^2 = 1. \tag{2}$$

Furthermore, exchanging the same two particles twice cannot change the wave function at all, hence

$$P_{ij}^2\,\psi(\vec{x}) = \lambda^2\psi(\vec{x}) = \psi(\vec{x}) \quad \Longrightarrow \quad \lambda^2 = 1, \text{ i.e. } \lambda = \pm 1. \tag{3}$$

Fermions are the particles for which $\lambda = -1$, i.e.

$$\boxed{\psi(\ldots,x_i,\ldots,x_j,\ldots) = -\,\psi(\ldots,x_j,\ldots,x_i,\ldots)} \tag{4}$$

and thus $\psi(\ldots,x_i,\ldots,x_i,\ldots) = 0$.

1.3 Slater determinants

Here we loosely follow E. Koch, *Mean-Field Theory: Hartree-Fock and BCS* (Jülich 2024), section 1.

Let $\{\varphi_n\}$ be a complete, orthonormal single-particle basis,

$$\sum_n \varphi_n^*(x)\varphi_n(x') = \delta(x-x') \qquad \text{(complete)}, \tag{5}$$

$$\int dx\,\varphi_n^*(x)\varphi_m(x) = \delta_{nm} \qquad \text{(orthonormal)}. \tag{6}$$

An arbitrary N-particle function can then be expanded in its first argument as

$$a(x_1,\ldots,x_N) = \sum_{n_1} a_{n_1}(x_2,\ldots,x_N)\,\varphi_{n_1}(x_1), \qquad \text{with} \tag{7}$$

$$a_{n_1}(x_2,\ldots,x_N) = \int dx_1\,\varphi_{n_1}^*(x_1)\,a(x_1,x_2,\ldots,x_N). \tag{8}$$

Iterating in all arguments,

$$\psi(x_1,\ldots,x_N) = \sum_{n_1,\ldots,n_N} a_{n_1,\ldots,n_N}\,\varphi_{n_1}(x_1)\cdots\varphi_{n_N}(x_N) \qquad \text{(not antisymmetric yet!)} \tag{9}$$

Now apply the antisymmetrizer $\mathcal{A}$, with $\mathrm{sgn}(P) = (-1)^P$:

$$\mathcal{A}\,\psi(x_1,\ldots,x_N) \equiv \frac{1}{\sqrt{N!}}\sum_P (-1)^P\,\psi(x_{P(1)},\ldots,x_{P(N)}) \tag{10}$$

$$= \frac{1}{\sqrt{N!}}\sum_P (-1)^P \sum_{n_1,\ldots,n_N} a_{n_1,\ldots,n_N}\,\varphi_{n_1}(x_{P(1)})\cdots\varphi_{n_N}(x_{P(N)}) \tag{11}$$

$$= \sum_{n_1,\ldots,n_N} a_{n_1,\ldots,n_N}\,\Phi_{n_1,\ldots,n_N}(x_1,\ldots,x_N), \tag{12}$$

where we have used the *Slater determinant*

$$\Phi_{\alpha_1,\ldots,\alpha_N}(x_1,\ldots,x_N) \equiv \frac{1}{\sqrt{N!}}\begin{vmatrix} \varphi_{\alpha_1}(x_1) & \varphi_{\alpha_2}(x_1) & \cdots & \varphi_{\alpha_N}(x_1) \\ \varphi_{\alpha_1}(x_2) & \varphi_{\alpha_2}(x_2) & \cdots & \varphi_{\alpha_N}(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ \varphi_{\alpha_1}(x_N) & \varphi_{\alpha_2}(x_N) & \cdots & \varphi_{\alpha_N}(x_N) \end{vmatrix} \tag{13}$$

$$\stackrel{\text{Leibniz}}{=} \frac{1}{\sqrt{N!}}\sum_{P\in S_N}\mathrm{sgn}(P)\prod_{i=1}^{N}\varphi_{\alpha_i}\big(x_{P(i)}\big). \tag{14}$$

Hence **any** $\psi(x_1,\ldots,x_N)$ **can be expanded in Slater determinants.** Furthermore, it follows immediately that $\psi(\vec{x})$ vanishes if either $x_i = x_j$ or $\alpha_i = \alpha_j$ for $i \neq j$: **fermions cannot occupy the same position** x **or the same quantum state** α**.**

1.4 The variational principle

Following F. Becca & S. Sorella, *Quantum Monte Carlo Approaches for Correlated Systems*, section 1.4.

Formulate some variational state $|\psi_{\rm var}\rangle$ and consider

$$E_{\rm var} = \frac{\langle\psi_{\rm var}|H|\psi_{\rm var}\rangle}{\langle\psi_{\rm var}|\psi_{\rm var}\rangle} \qquad \text{[B\&S (1.38)]}. \tag{15}$$

Expanding in the energy eigenstates $|\varphi_i\rangle$,

$$|\psi_{\rm var}\rangle = \sum_i |\varphi_i\rangle\langle\varphi_i|\psi_{\rm var}\rangle = \sum_i a_i|\varphi_i\rangle \qquad \text{[B\&S (1.39)]}, \tag{16}$$

so that, using $\langle\varphi_i|\varphi_j\rangle = \delta_{ij}$,

$$E_{\rm var} = \Big(\sum_i |a_i|^2\Big)^{-1}\Big(\sum_i a_i|\varphi_i\rangle\Big)^{\dagger} H\Big(\sum_j a_j|\varphi_j\rangle\Big) = \Big(\sum_i |a_i|^2\Big)^{-1}\sum_i |a_i|^2 E_i. \tag{17}$$

Assume $E_i \leq E_j$ for $i < j$, i.e. E_0 is the ground-state energy. Then

$$E_{\rm var} - E_0 = \underbrace{\Big(\sum_i |a_i|^2\Big)^{-1}}_{>0}\Big(\sum_{i\neq 0}\underbrace{|a_i|^2}_{\geq 0}\underbrace{(E_i - E_0)}_{\geq 0}\Big) \;\geq\; 0 \qquad \text{[B\&S (1.41)]}, \tag{18}$$

i.e. $E_{\rm var}$ **is an upper bound for the exact energy.** In practice: parametrize some $|\psi_{\rm var}\rangle$ and minimize $E_{\rm var}$.

1.5 Variational wave functions

1.5.1 Hartree-Fock wave functions

The Hartree-Fock (HF) approximation writes the many-body wave function as a product of single-particle orbitals,

$$\boxed{|\psi_{\mathrm{HF}}\rangle = \prod_{\alpha=1}^{N_e} \phi_\alpha^\dagger |0\rangle, \qquad \phi_\alpha^\dagger = \sum_{i=1}^{N}\sum_{\sigma} W_{\sigma,\alpha,i}\, c_{i,\sigma}^\dagger, \quad \sigma = \uparrow,\downarrow} \tag{19}$$

with N lattice sites and N_e electrons. This leaves $2 \cdot N_e \cdot N$ parameters instead of the exponentially many amplitudes of a generic many-body state. The parameters are then optimized by minimizing

$$E_{\mathrm{HF}} = \frac{\langle\psi_{\mathrm{HF}}|H|\psi_{\mathrm{HF}}\rangle}{\langle\psi_{\mathrm{HF}}|\psi_{\mathrm{HF}}\rangle}, \qquad \text{e.g.} \quad W_{\sigma,\alpha,i} \;\to\; W_{\sigma,\alpha,i} - \frac{\partial E_{\mathrm{HF}}}{\partial W^*_{\sigma,\alpha,i}}\,\delta\tau \qquad \text{[B\&S (1.54)]}, \tag{20}$$

using Wirtinger derivatives.

The HF wave function can equivalently be written in first quantization as **one Slater determinant**. Since it is a *product of single-particle orbitals*, it is mostly useful in the non-interacting case or at weak coupling. In the Hubbard model, turning up U makes the motion of one electron depend on where the other electrons are, i.e. it generates electron-electron correlation.

1.5.2 The Gutzwiller wave function

Gutzwiller's idea: describe the effect of the Hubbard U interaction by reducing the weight of configurations with multiply occupied sites. Start from some ground state $|\psi_0\rangle$ (e.g. non-interacting, a Slater determinant) and apply an operator P_g that suppresses such configurations:

$$\boxed{|\psi_g\rangle = P_g|\psi_0\rangle} \qquad \text{with} \qquad \boxed{P_g = \exp\Big[-\frac{g}{2}\sum_{i=1}^{N}(n_i - n)^2\Big]} \qquad \text{[B\&S (1.58),(1.59)]}, \tag{21}$$

where $n_i = \sum_\sigma n_{i\sigma}$ and $n = \frac{1}{N}\big\langle \sum_{i=1}^{N} n_i \big\rangle$ is the average density per site. Here g is the variational parameter ($g > 0$ for the repulsive Hubbard model).

Given some configuration $|x\rangle$, P_g counts how many sites within that state have higher or lower occupation than the average; for repulsive interactions these states are suppressed accordingly:

$$\langle x|\psi_g\rangle = P_g(x)\,\langle x|\psi_0\rangle, \qquad P_g(x) \le 1 \qquad \text{[B\&S (1.60)]}. \tag{22}$$

Since $P_g(x) \ge 0$, it <u>does not</u> affect the sign structure.

In the Hubbard model a metal-insulator transition is expected at *finite* U/t for $n = 1$. The Gutzwiller wave function can only reproduce this for $g \to \infty$; otherwise there is always a non-zero probability for doublon-holon creation, and these could then move around freely and thus conduct.

Fully projected Gutzwiller wave function ($g \to \infty$). Examine P_∞ for $n = 1$:

$$P_\infty = \lim_{g\to\infty} P_g = \lim_{g\to\infty} \exp\Big[-\frac{g}{2}\sum_{i=1}^{N}(\hat{n}_i - 1)^2\Big], \tag{23}$$

$$\Rightarrow \quad P_\infty|\psi\rangle = \lim_{g\to\infty}\begin{cases}\exp\big(-\frac{g}{2}a\big), & \text{if } \exists \text{ one site with a holon or doublon } (a>0)\\ \exp(0), & \text{else}\end{cases} \quad |\psi\rangle = \begin{cases}0\\ |\psi\rangle\end{cases} \tag{24}$$

Therefore we can rewrite P_∞ as

$$\boxed{P_\infty = \prod_i \big(n_{i,\uparrow} - n_{i,\downarrow}\big)^2} \qquad \text{[B\&S (1.63)]}. \tag{25}$$

If we want to allow empty sites, one has to set

$$\boxed{P_\infty = \prod_i \big(1 - n_{i,\uparrow}n_{i,\downarrow}\big)}, \tag{26}$$

which is important for $n < 1$, since then holons and doublons do not come in pairs.

For $g = \infty$ no density fluctuations can occur, whereas in a true insulator density fluctuations do occur but produce no net current – e.g. double-occupancy fluctuations via virtual processes in the Hubbard model at large U (see Fazekas, chapter 5, t-J model).

1.5.3 The density-density Jastrow wave function

Since the Gutzwiller wave function does not correctly describe an insulator, the ansatz requires modification. We can, for instance, change the correlation term that is applied to $|\psi_0\rangle$ (or change $|\psi_0\rangle \rightsquigarrow |\psi_{U=\infty}\rangle$). The Jastrow factor does this by including long-range correlations:

$$\boxed{\mathcal{J} = \exp\Big[-\frac{1}{2}\sum_{i,j} v_{ij}\,(n_i - n)(n_j - n)\Big]} \qquad \text{[B\&S (1.65)]} \tag{27}$$

$$\boxed{|\psi_\mathcal{J}\rangle = \mathcal{J}|\psi_0\rangle = |\psi_g\rangle \quad \text{for } v_{ij} = \delta_{ij}\, g} \qquad \text{[B\&S (1.64)]} \tag{28}$$

For translationally invariant systems the variational parameters are $v_{ij} = v_{|i-j|}$, i.e. $\mathcal{O}(L)$ possible different distances.

The long-range correlations induced by $\mathcal{J}$ shall create a bound state between holons and doublons, possibly hindering conduction while still allowing local density fluctuations. It can be shown that the behaviour of v_{ij} resulting from an energy minimization is <u>not</u> enough to capture a fully gapped phase (power-law behaviour is present, but the Fourier transform of v_{ij} decays exponentially).

Jastrow-Slater wave function. In the case $v_{ij} = v_{ji}$ the Jastrow factor is symmetric under particle exchange, so (in first quantization) it can be combined with a Slater determinant to obtain a suitable fermionic wave function, $\psi_{\rm JS}(\vec{x}) = \langle \vec{x}|\psi_{\rm JS}\rangle$:

$$\psi_{\rm JS}(\vec{r}_1, \ldots, \vec{r}_{N_e}) = \exp\Big[-\sum_{i<j} v_{ij}(n_i - n)(n_j - n)\Big] \cdot \psi_{\rm HF}(\vec{r}_1, \ldots, \vec{r}_{N_e}) \qquad \text{[B\&S} \sim\text{(1.66)]}, \tag{29}$$

with $\psi_{\rm HF}(\vec{r}_1, \ldots, \vec{r}_{N_e}) = \det\{\phi_\alpha(\vec{r}_i)\}$ and $\{\phi_\alpha(\vec{r}_i)\}$ a set of one-particle orbitals.

1.5.4 The backflow wave function

As explained earlier, the set of Slater determinants forms a basis of the N-electron Hilbert space, so one can systematically improve HF by adding more Slater determinants – but this is computationally very demanding. Instead: **try to improve one single Slater determinant by making the orbitals configuration dependent**, which goes beyond mean field.

Original idea. The movement of one particle "pushes other particles away" in order to avoid large overlap among them, i.e. a "counter flow" of all other particles – hence *backflow correlations*. One therefore constructs the Slater determinant with modified coordinates (continuum backflow),

$$\vec{r}_i^{\,b} = \vec{r}_i + \sum_{j\neq i} \eta\big(|\vec{r}_i - \vec{r}_j|\big)\big(\vec{r}_i - \vec{r}_j\big), \tag{30}$$

which also describes the effective displacement of the i-th particle due to the j-th one.

On a lattice. For lattice models this effective displacement is much less useful, since all distances are discretized. Instead one changes the one-particle orbitals:

$$\boxed{\phi_k\big(\vec{r}^{\,b}_{i,\sigma}\big) \;\simeq\; \phi_k^b\big(\vec{r}_{i,\sigma}\big) \equiv \phi_k\big(\vec{r}_{i,\sigma}\big) + \sum_j c_{ij}[x]\,\phi_k\big(\vec{r}_{j,\sigma}\big)} \tag{31}$$

where $\phi_k(\vec{r}_{i,\sigma}) = \langle 0|c_{i,\sigma}|\phi_k\rangle$, the $|\phi_k\rangle$ are eigenstates of the mean-field Hamiltonian, and $|x\rangle$ is the many-body configuration of the system.

This is essentially a *dynamic linear combination of orbitals*: dynamic, since it *depends on the electron configuration.* In the Hubbard model at large U one can hybridize the orbitals as follows:

$$\boxed{\phi_k^b(\vec{r}_{i,\sigma}) = \varepsilon\,\phi_k(\vec{r}_{i,\sigma}) + \eta\sum_j t_{ij}\,\big(D_i H_j\big)\,\phi_k(\vec{r}_{j,\sigma})} \tag{32}$$

i.e. hybridize the orbitals at i and j if there is a doublon at i and a holon at j, such that the electron can minimize energy by going to the empty site (cf. Tocchio, Sorella et al., Eqs. (2)-(4)).

Summary. Backflow correlations improve a single Slater determinant by replacing fixed single-particle orbitals with **configuration-dependent orbitals**, effectively introducing **many-body corrections** through a determinant whose elements depend on the full configuration. Combining this with a Jastrow factor gives the Slater-Jastrow-backflow wave function.

1.6 Parametrization via neural networks

Now let us have a look at how neural networks are used for variational states. There is no single resource that reviews the physics-motivated approaches like backflow, hidden fermions etc. – so we mostly look at the corresponding papers.

1.6.1 Backflow transformations via neural networks

Following D. Luo & B.K. Clark, *Backflow Transformations via Neural Networks for Quantum Many-Body Wave Functions*.

Recall the definition of backflow introduced by Tocchio et al. in "Role of backflow correlations for the nonmagnetic phase of the t-t' Hubbard model", Eq. (4):

$$\phi^b_{k\sigma}\big(r_{i,\sigma};\vec{x}\big) = \phi_{k\sigma}\big(r_{i,\sigma}\big) + \sum_j \eta_{ij,\sigma}\,\phi_{k\sigma}\big(r_{j,\sigma}\big), \tag{33}$$

$$\eta_{ij,\sigma} = t\,D_i H_j\,\Theta_{|i-j|,\sigma}, \tag{34}$$

with $D_i = n_{i,\uparrow}n_{i,\downarrow}$ and $H_i = (1-n_{i,\uparrow})(1-n_{i,\downarrow})$ counting doublons and holons.

The i,j-dependence of the hopping amplitudes t_{ij} is absorbed in the variational parameters $\Theta_{|i-j|,\sigma}$; e.g. for next-nearest-neighbour hopping one sets them to zero if $|i-j|\neq 1,2$.

Now we make the ansatz to parametrize the backflow part via a neural network:

$$\boxed{\phi^b_{k\sigma}\big(r_{i,\sigma};\vec{x}\big) = \phi_{k\sigma}\big(r_{i,\sigma}\big) + a^{\mathrm{NN}}_{ki,\sigma}(\vec{x})} \tag{35}$$

The typical advantage of a NN parametrization is that neural networks are universal approximators. On top of that, one can show **explicitly how to construct the backflow with a NN with two hidden layers**:

Step 1: rewrite η as a ReLU. Assuming $t\,\Theta_{|i-j|,\sigma} \geq 0$,

$$\eta_{ij,\sigma} = t\,D_i H_j\,\Theta_{|i-j|,\sigma} = t\,n_{i,\uparrow}n_{i,\downarrow}h_{j,\uparrow}h_{j,\downarrow}\,\Theta_{|i-j|,\sigma}, \tag{36}$$

which is nonzero only if $n_{i,\uparrow} = n_{i,\downarrow} = h_{j,\uparrow} = h_{j,\downarrow} = 1$, and hence

$$\eta_{ij,\sigma} = \mathrm{ReLU}\Big[t\,\Theta_{|i-j|,\sigma}\big(n_{i,\uparrow}+n_{i,\downarrow}+h_{j,\uparrow}+h_{j,\downarrow}-3\big)\Big]. \tag{37}$$

Step 2: input layer. The NN shall have the input layer

$$\big(\underbrace{\sigma_1,\ldots,\sigma_N}_{\text{spin up}},\underbrace{\sigma_{N+1},\ldots,\sigma_{2N}}_{\text{spin down}}\big), \qquad \sigma_i = 2n_{i,\uparrow}-1, \quad \sigma_{i+N} = 2n_{i,\downarrow}-1, \quad i=1,\ldots,N. \tag{38}$$

Inverting the relation for σ_i and inserting it into $\eta_{ij,\sigma}$,

$$\eta_{ij,\sigma} = \mathrm{ReLU}\Big[\tfrac{1}{2}\,t\,\Theta_{|i-j|,\sigma}\big(\sigma_i+\sigma_{i+N}-\sigma_j-\sigma_{j+N}-2\big)\Big], \qquad i,j\in\{1,\ldots,N\},\ \sigma=\pm1. \tag{39}$$

Step 3: first hidden layer. Use this to construct $\eta_{ij,\sigma}$ as the following NN:

$$v_k = \sum_{\ell=1}^{2N} W_{k\ell}\, \sigma_\ell + b_k, \qquad \text{set} \quad k = \sigma \cdot (i + N \cdot j). \tag{40}$$

$$\Rightarrow \qquad v_{\sigma\cdot(i+N\cdot j)} = \sum_{\ell=1}^{2N} W_{\sigma\cdot(i+N\cdot j),\ell}\, \sigma_\ell + b_{\sigma\cdot(i+N\cdot j)}, \tag{41}$$

with

$$b_{\sigma\cdot(i+N\cdot j)} = -t\,\Theta_{|i-j|,\sigma} \qquad \text{and} \qquad W_{\sigma\cdot(i+N\cdot j),\ell} = \frac{t}{2}\,\Theta_{|i-j|,\sigma} \begin{cases} 1 & \text{if } \ell \,\%\, N = i \\ -1 & \text{if } \ell \,\%\, N = j \\ 0 & \text{else} \end{cases} \tag{42}$$

such that $\eta_{ij,\sigma} = \mathrm{ReLU}\big(v_{\sigma\cdot(i+N\cdot j)}\big)$.

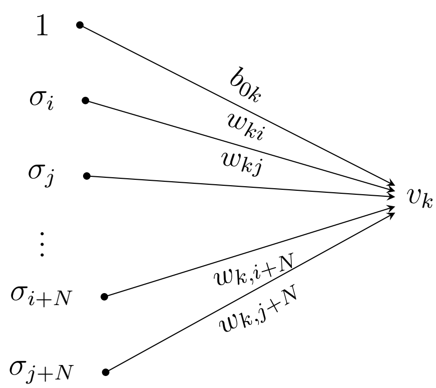

Figure 1: First hidden layer producing v_k, with $k = \sigma \cdot (i + N \cdot j)$.

Step 4: second hidden layer. Finally, the full backflow is obtained by adding another hidden layer which takes the $\eta_{ij,\sigma}$ as inputs and weights them by the single-particle orbitals $\phi_k(r_{j,\sigma})$. The activation function of this layer is simply the identity. If $t\,\Theta_{|i-j|,\sigma} < 0$, the minus sign can be absorbed into this layer, resulting in

$$a^{\mathrm{NN}}_{ki,\sigma}(\vec{r}) = \sum_j \underbrace{\eta_{ij,\sigma}}_{\text{inputs}}\, \underbrace{s_{ij,\sigma}\,\phi_{k\sigma}(r_{j,\sigma})}_{\text{weights}}, \qquad s_{ij,\sigma} = \mathrm{sgn}\big(t\,\Theta_{|i-j|,\sigma}\big) = \pm 1. \tag{43}$$

$\Rightarrow$ **the backflow can be represented by a neural network.**

In practice, one simply chooses some NN (RBM, MLP, CNN, ...) and lets backpropagation take care of the rest.

1.7 References

- F. Schwabl, *Quantum Mechanics*, section 13.1.

- E. Koch, *Mean-Field Theory: Hartree-Fock and BCS*, Jülich lecture notes (2024), section 1.
- F. Becca & S. Sorella, *Quantum Monte Carlo Approaches for Correlated Systems*, sections 1.4-1.5.
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- L.F. Tocchio, F. Becca, A. Parola & S. Sorella, *Role of backflow correlations for the nonmagnetic phase of the t-t' Hubbard model*, Phys. Rev. B **78**, 041101(R) (2008).
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2 Similarity transformations of operators: computing $e^A B e^{-A}$

2.1 Motivation

Similarity transformations $B \mapsto e^A B e^{-A}$ are ubiquitous whenever a Hamiltonian is brought to a simpler (often diagonal) form: Bogoliubov transformations of quadratic Hamiltonians, displacement/squeezing of bosonic modes, polaron (Lang-Firsov) transformations, Schrieffer-Wolff transformations, and passage to the interaction picture or a rotating frame are all instances of this construction. If $A^\dagger = -A$ (generator anti-Hermitian), $U = e^A$ is unitary and the transformation preserves Hermiticity and spectra. This is the case in all examples below.

2.2 Method A: The Hadamard lemma

Hadamard Lemma. *For any two operators A, B,*

$$e^A B e^{-A} = \sum_{n=0}^{\infty} \frac{1}{n!} \operatorname{ad}_A^n(B), \qquad \textit{with} \tag{44}$$

$$\operatorname{ad}_A^n(B) \equiv [A, B]_n \equiv [A, [A, B]_{n-1}], \quad \operatorname{ad}_A^0(B) \equiv [A, B]. \tag{45}$$

This series is especially helpful whenever the nested commutator yields zero at for a finite n or if the nested commutators eventually reproduce a small, recognizable pattern.

Strategy 1: Reduce to elementary operators. Utilize that for any A, X, Y,

$$e^A (XY) e^{-A} = \left(e^A X e^{-A}\right)\left(e^A Y e^{-A}\right). \tag{46}$$

It thus suffices to compute $e^A\, a\, e^{-A}$ and $e^A\, a^\dagger e^{-A}$ once and any polynomial in these follows by substitution and normal-ordering, with no further Hadamard series needed.

Strategy 2: The dagger shortcut. If $A^\dagger = -A$, $U = e^A$ is unitary with $U^\dagger = e^{-A} = U^{-1}$, so $e^A B e^{-A} = U B U^\dagger$. Therefore, taking the dagger simply yields

$$\left(e^A B e^{-A}\right)^\dagger = e^A B^\dagger e^{-A}. \tag{47}$$

Once $e^A a e^{-A}$ is known, $e^A a^\dagger e^{-A}$ follows for free by Hermitian conjugation, roughly halving the bookkeeping.

2.3 Method B: The generator flow method (not proof read since not used yet)

Define $X(\theta) \equiv e^{\theta A}\, X\, e^{-\theta A}$, $X(0) = X$. Differentiating,

$$\frac{dX(\theta)}{d\theta} = [A, X(\theta)] = \operatorname{ad}_A\big(X(\theta)\big). \tag{48}$$

This ordinary differential equation (ODE) is exactly equivalent to Method A – the Hadamard series is its formal Taylor solution around $\theta = 0$ – but is can be more tractable in practice.

Key fact: closure $\Rightarrow$ finite-dimensional linear ODE. If ad_A maps a finite set of operators $\{X_1, \ldots, X_n\}$ into linear combinations of themselves, the operator ODE collapses to a matrix ODE:

$$\frac{d}{d\theta}\vec{X}(\theta) = M\,\vec{X}(\theta), \qquad [A, X_j] = \sum_i M_{ij} X_i, \qquad \vec{X}(\theta) = e^{\theta M}\vec{X}(0), \tag{49}$$

reducing the problem to exponentiating an ordinary $n \times n$ matrix.

2.4 When to prefer which method

Both methods are mathematically identical. Method A is fast when the pattern is recognizable by eye; Method B is more systematic for anything beyond the simplest single-operator case, especially with mixed or multiple generators to track jointly.

2.5 Scope: when this machinery fails

The closure property requires A to be at most quadratic in a set of canonically (anti-) commuting operators. For a genuinely quartic operator such as $H_{\text{int}} = U \sum_i (n_i - \frac{1}{2})(n_{i+1} - \frac{1}{2})$, $\mathrm{ad}_A(H_{\text{int}})$ does not close on a finite set for generic one-body A (e.g. $A \propto J$ or $A \propto T$) – $[J, H_{\text{int}}]$ is generically nonzero and does not reduce to a small closed set of operators. The Hadamard series then cannot be resummed in closed form; one is limited to a perturbative treatment in the strength of A.

2.6 References

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3 The tensor product and the Kronecker product as one of its possible representations

3.1 The tensor product of vector spaces

Let $\underline{v} = \{v_i \mid i \in I\}$ and $\underline{w} = \{w_j \mid j \in J\}$ be bases for V and W. Then $V \otimes W$ is the product space, where there exists a basis which can be identified uniquely with the pairs of the cartesian product

$$\underline{v} \times \underline{w} = \big\{(v_i, w_j) \mid i \in I,\ j \in J\big\}, \tag{50}$$

so that

$$\dim(V \otimes W) = \dim V \cdot \dim W. \tag{51}$$

Since $V \otimes W$ is a vector space, an arbitrary element of it can be written as

$$\sum_{(i,j)\in I\times J} c_{ij}\,(v_i \otimes w_j). \tag{52}$$

The bilinear map $\otimes$ is defined by

$$\otimes\colon V \times W \longrightarrow V \otimes W, \tag{53}$$

$$(v, w) \longmapsto v \otimes w = \sum_{(i,j)\in I\times J} a_i b_j\,(v_i \otimes w_j), \tag{54}$$

for $v = \sum_i a_i v_i$ and $w = \sum_j b_j w_j$. As a check of bilinearity, writing $v' = \sum_i a_i' v_i$ and $v'' = \sum_i a_i'' v_i$,

$$\begin{aligned}(v' + v'') \otimes w &= \Big(\textstyle\sum_i a_i' v_i + \sum_i a_i'' v_i\Big) \otimes w = \Big(\textstyle\sum_i (a_i' + a_i'')\, v_i\Big) \otimes w \\ &= \sum_{i,j} (a_i' + a_i'')\, b_j\,(v_i \otimes w_j) \\ &= \sum_{i,j} a_i' b_j\,(v_i \otimes w_j) + \sum_{i,j} a_i'' b_j\,(v_i \otimes w_j) \\ &= v' \otimes w + v'' \otimes w,\end{aligned} \tag{55}$$

and analogously in the second argument.

3.2 The tensor product of operators

Introduce operators $A\colon V \to V$ and $B\colon W \to W$ with V, W vector spaces. How is $A \otimes B\colon V \otimes W \to V \otimes W$ given? Define a map $\beta\colon V \times W \to V \otimes W$, $\beta(v, w) := Av \otimes Bw$. It is bilinear, e.g.

$$\begin{aligned}\beta(v_1 + v_2, w) &= A(v_1 + v_2) \otimes Bw = (Av_1 + Av_2) \otimes Bw \\ &\overset{\otimes\ \text{bilin.}}{=} Av_1 \otimes Bw + Av_2 \otimes Bw = \beta(v_1, w) + \beta(v_2, w),\end{aligned} \tag{56}$$

and analogously in the second argument. By the universal property there exists a unique linear $\tilde\beta$ with $\tilde\beta(v \otimes w) := \beta(v, w)$, so that

$$(A \otimes B)(v \otimes w) = Av \otimes Bw. \tag{57}$$

3.3 The Kronecker product as a matrix representation

Let $V = K^m$ and $W = K^n$ and identify (this is *not* a unique representation!) $K^m \times K^n \to K^{(m \cdot n)}$ with the Kronecker product, i.e. for

$$v = \begin{pmatrix} a_1 \\ \vdots \\ a_m \end{pmatrix} \quad \& \quad w = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \qquad \text{set} \qquad v \otimes w := \begin{pmatrix} a_1\, w \\ \vdots \\ a_m\, w \end{pmatrix} = \begin{pmatrix} a_1 b_1 \\ \vdots \\ a_1 b_n \\ \vdots \\ a_m b_1 \\ \vdots \\ a_m b_n \end{pmatrix}. \tag{58}$$

Now choose $\{v_i\}$ to be the standard basis, so that $Av_i = A_{\cdot i}$ (the i-th column of A) and likewise for B. Then

$$\begin{aligned} (A \otimes B)(v_i \otimes v_j) = (Av_i) \otimes (Bv_j) &\overset{\text{std. basis}}{=} A_{\cdot i} \otimes B_{\cdot j} \\ &= \begin{pmatrix} A_{1i}\, B_{\cdot j} \\ \vdots \\ A_{mi}\, B_{\cdot j} \end{pmatrix} = \begin{pmatrix} \begin{pmatrix} A_{1i} B_{1j} \\ \vdots \\ A_{1i} B_{nj} \end{pmatrix} \\ \vdots \\ \begin{pmatrix} A_{mi} B_{1j} \\ \vdots \\ A_{mi} B_{nj} \end{pmatrix} \end{pmatrix}. \end{aligned} \tag{59}$$

Setting $v_i \otimes v_j \equiv w_k$ with $k = j + n\, i$ (for zero-based i), the matrix of $A \otimes B$ is the block matrix

$$\begin{aligned} A \otimes B &= \begin{pmatrix} A_{11} B_{\cdot 1} & \cdots & A_{11} B_{\cdot n} & \cdots & A_{1m} B_{\cdot 1} & \cdots & A_{1m} B_{\cdot n} \\ \vdots & & & & & & \vdots \\ A_{m1} B_{\cdot 1} & \cdots & A_{m1} B_{\cdot n} & \cdots & A_{mm} B_{\cdot 1} & \cdots & A_{mm} B_{\cdot n} \end{pmatrix} \\ &= \begin{pmatrix} A_{11} B & \cdots & A_{1m} B \\ \vdots & \ddots & \vdots \\ A_{m1} B & \cdots & A_{mm} B \end{pmatrix}, \end{aligned} \tag{60}$$

which is the Kronecker product for matrices.

3.4 References

- *Tensorprodukt* and *Kronecker-Produkt*, German Wikipedia.

4 The Jordan–Wigner mapping for spinless Fermions

4.1 Summary

The Jordan–Wigner transformation (JWT) can be used to map a fermionic system to a system of spins. This is essentially done by rewriting

$$\text{fermion} = \text{string} \times \text{operator},$$

where the "string" is some product of operators $\hat{Z}_i = 1 - 2\hat{n}_i$ and the "operator" is found to obey spin commutation relations.

As references I used

- *Fermions and Jordan–Wigner String* – ITensor C++ Documentation,
- *Introduction to Many-Body Physics*, Chapter 4.1 – Piers Coleman,
- *The Fermionic canonical commutation relations and the Jordan–Wigner transform* – M. Nielsen,

which I supplemented with some calculations and my own way of structuring things.

4.2 Setup: the fermionic Hilbert space

Our fermionic Hilbert space is defined by (Nielsen):

i) operators $c_1, \ldots, c_L$ acting on some Hilbert space;

ii) these operators satisfying the canonical (anti-)commutation relations for fermions

$$\{c_i^{(+)}, c_j^{(+)}\} = 0 \qquad \& \qquad \{c_i, c_j^\dagger\} = \delta_{ij}\,\mathbb{1}. \tag{61}$$

Many things directly follow from this, such as a possible basis for the Hilbert space, namely the occupation number basis

$$|\vec{n}\rangle = |n_1, n_2, \ldots, n_L\rangle \equiv \left(\prod_{i=1}^{L} (c_i^\dagger)^{n_i}\right)|0\rangle\,, \tag{62}$$

where $\hat{n}_i\,|\vec{n}\rangle = c_i^\dagger c_i\,|\vec{n}\rangle = n_i\,|\vec{n}\rangle$ with $n_i = 0, 1$ and

$$[\hat{n}_i, \hat{n}_j] = 0. \tag{63}$$

4.3 The transformation

The JWT introduces new operators a_j by writing

$$c_j \equiv \left(\prod_{i=1}^{j-1} Z_i\right) a_j \qquad \& \qquad c_j^\dagger \equiv a_j^\dagger \left(\prod_{i=1}^{j-1} Z_i\right), \tag{64}$$

with $Z_j = (-1)^{n_j} = 1 - 2n_j$ and $n_j = c_j^\dagger c_j$. The ordering of the Z_i does not matter since $[Z_i, Z_j] = 0$ due to eq. (63). Inverting eq. (64) using $Z_i^2 = \mathbb{1}$ and $[Z_i, Z_j] = 0$ yields

$$a_j = \left(\prod_{i=1}^{j-1} Z_i\right) c_j \qquad \& \qquad a_j^\dagger = c_j^\dagger \left(\prod_{i=1}^{j-1} Z_i\right). \tag{65}$$

4.4 Algebra of the new operators

One can show that these new operators obey

$$[a_i^{(+)}, a_j^{(+)}] = 0 \qquad \forall\, i, j, \tag{66}$$

$$[a_i, a_j^\dagger] = 0 \qquad \forall\, i \neq j, \tag{67}$$

$$[a_i, a_i^\dagger] = 1 - 2n_i \qquad \text{with } n_i = c_i^\dagger c_i, \tag{68}$$

using that

$$\{Z_i, c_j^{(+)}\} = 0 \tag{69}$$

The corresponding proofs can be found in section 4.8.

4.5 The occupation-number basis in the new operators

The basis states of the fermionic Hilbert space can be written as

$$\begin{aligned}
|n_1, n_2, \ldots, n_L\rangle &\equiv \left(\prod_{i=1}^{L} (c_i^\dagger)^{n_i}\right) |0\rangle && \text{w. } n_i = 0, 1 \\
&= \left\{\prod_{i=1}^{L} \left[a_i^\dagger \left(\prod_{j=1}^{i-1} Z_j\right)\right]^{n_i}\right\} |0\rangle && (c_1^\dagger = a_1^\dagger) \\
&= (a_1^\dagger)^{n_1} (a_2^\dagger Z_1)^{n_2} (a_3^\dagger Z_2 Z_1)^{n_3} \cdots (a_L^\dagger Z_{L-1} \cdots Z_1)^{n_L} |0\rangle \\
&= \left(\prod_{i=1}^{L} (a_i^\dagger)^{n_i}\right) (Z_1)^{n_2} (Z_2 Z_1)^{n_3} \cdots (Z_{L-1} \cdots Z_1)^{n_L} |0\rangle \\
&= \left(\prod_{i=1}^{L} (a_i^\dagger)^{n_i}\right) |0\rangle && \text{w. } n_i = 0, 1,
\end{aligned} \tag{70}$$

using $[Z_i, Z_j] = 0\ \forall i, j$, $[Z_i, c_j] = 0$ for $i \neq j$ (hence $[Z_i, a_j^\dagger] = 0$ for $i \neq j$), and $Z_j |0\rangle = |0\rangle$. The ordering is not important anymore because $[a_i^\dagger, a_j^\dagger] = 0$.

4.6 Action on basis states

One cannot define the action of a_j ($a_j^\dagger$) locally, since the JW string $Z_1 \cdots Z_{j-1}$ is non-local, so one has to examine them on the full many-body basis. For $n_\ell = 0$,

$$\begin{aligned} a_\ell \left| n_1, \dots, n_\ell = 0, \dots, n_L \right\rangle &= a_\ell \left(\prod_{i=1}^{L} (a_i^\dagger)^{n_i} \right) \left| 0 \right\rangle \\ &\overset{[a_\ell, a_j^\dagger]=0}{=} \left(\prod_{i=1}^{L} (a_i^\dagger)^{n_i} \right) a_\ell \left| 0 \right\rangle = 0, \end{aligned} \tag{71}$$

since $a_\ell \left| 0 \right\rangle = c_\ell \left| 0 \right\rangle = 0$. For $n_\ell = 1$,

$$\begin{aligned} a_\ell \left| n_1, \dots, n_\ell = 1, \dots, n_L \right\rangle &= a_\ell \left(\prod_{\substack{i=1 \\ i \neq \ell}}^{L} (a_i^\dagger)^{n_i} \right) a_\ell^\dagger \left| 0 \right\rangle \\ &\overset{[a_\ell, a_j^\dagger]=0}{=} \left(\prod_{\substack{i=1 \\ i \neq \ell}}^{L} (a_i^\dagger)^{n_i} \right) a_\ell a_\ell^\dagger \left| 0 \right\rangle = \left(\prod_{\substack{i=1 \\ i \neq \ell}}^{L} (a_i^\dagger)^{n_i} \right) c_\ell c_\ell^\dagger \left| 0 \right\rangle \\ &= \left(\prod_{\substack{i=1 \\ i \neq \ell}}^{L} (a_i^\dagger)^{n_i} \right) \left| 0 \right\rangle = \left| n_1, \dots, n_\ell = 0, \dots, n_L \right\rangle . \end{aligned} \tag{72}$$

Using $[a_\ell^\dagger, a_j^\dagger] = 0$ and $(a_\ell^\dagger)^2 = 0$, one quickly finds

$$a_\ell^\dagger \left| n_1, \dots, n_\ell = 0, \dots, n_L \right\rangle = \left| n_1, \dots, n_\ell = 1, \dots, n_L \right\rangle , \tag{73}$$

$$a_\ell^\dagger \left| n_1, \dots, n_\ell = 1, \dots, n_L \right\rangle = 0. \tag{74}$$

Thus a_ℓ ($a_\ell^\dagger$) destroys (creates) a particle in the many-body state *without* giving rise to an extra sign structure (as c_ℓ ($c_\ell^\dagger$) would do).

4.7 Identification with spins

We can thus identify $n_i = 0, 1$ with $n_i = \downarrow, \uparrow$ in the many-body state, and the new operators can be associated with spins by identifying

$$a_j = S_j^-, \qquad a_j^\dagger = S_j^+ \qquad \& \qquad n_j - \tfrac{1}{2} = S_j^z = \tfrac{1}{2}\,\sigma_j^z, \tag{75}$$

since $[S_j^-, S_j^+] = -2S_j^z$ for spins (see the quick refresher on spin, section 7). For instance

$$a_3^\dagger \left| 0 \right\rangle = S_3^+ \left| \downarrow\downarrow\downarrow \right\rangle = \left| \uparrow\downarrow\downarrow \right\rangle .$$

Matrix elements do not change under the JWT, since it is only a rewriting of operators and a renaming of basis states.

4.8 Proofs about the JWT

Claim: $\{Z_j, c_j^{(+)}\} = 0$**.**

Proof. On the single site j, with $Z_j |0\rangle_j = |0\rangle_j$, $Z_j |1\rangle_j = -|1\rangle_j$, $c_j |0\rangle_j = 0$ and $c_j |1\rangle_j = |0\rangle_j$,

$$Z_j c_j |0\rangle_j = Z_j \cdot 0 = 0 = c_j |0\rangle_j = c_j Z_j |0\rangle_j \,, \tag{76}$$

$$Z_j c_j |1\rangle_j = Z_j |0\rangle_j = |0\rangle_j = c_j |1\rangle_j \,, \tag{77}$$

$$c_j Z_j |1\rangle_j = -c_j |1\rangle_j = -|0\rangle_j \,. \tag{78}$$

Hence, on both basis states $\{Z_j, c_j\} = Z_j c_j + c_j Z_j = c_j - c_j = 0$, and since $Z_j^\dagger = Z_j$ it follows that $\{Z_j, c_j^\dagger\} = 0$ as well. (see Coleman, Chapter 4.1) □

Claim: $[a_i^{(+)}, a_j^{(+)}] = 0 \ \ \forall\, i, j$**.**

Proof. Assume w.l.o.g. that $i < j$. Using eq. (65),

$$\begin{aligned} a_j a_i &= (Z_1 \cdots Z_{j-1} c_j)\,(Z_1 \cdots Z_{i-1} c_i) \\ &\overset{\substack{[Z_j, c_\ell]=0,\, j\neq\ell \\ \{c_i, c_j\}=0}}{=} Z_1^2 \cdots Z_{i-1}^2\, Z_i \cdots Z_{j-1}\, c_j c_i = -Z_i \cdots Z_{j-1}\, c_i c_j \qquad (= 0 \text{ if } i = j), \end{aligned} \tag{79}$$

and likewise

$$\begin{aligned} a_i a_j &= (Z_1 \cdots Z_{i-1} c_i)\,(Z_1 \cdots Z_{j-1} c_j) \\ &\overset{[Z_i, c_\ell]=0,\, i\neq\ell}{=} Z_{i+1} \cdots Z_{j-1}\, c_i Z_i c_j \overset{\{Z_i, c_i\}=0}{=} -Z_i \cdots Z_{j-1}\, c_i c_j. \end{aligned} \tag{80}$$

Comparing the two, $a_i a_j = a_j a_i$, i.e. $[a_i, a_j] = 0$. Analogously for $[a_i^\dagger, a_j^\dagger]$ and for $i > j$; the case $i = j$ is straightforward since $c_i^2 = 0$. □

Claim: $[a_i, a_j^\dagger] = 0 \ \ \forall\, i \neq j$**.**

Proof. We can show it w.l.o.g. for $i < j$. Using eq. (65) and eq. (69),

$$\begin{aligned} a_i^\dagger a_j &= \big(c_i^\dagger Z_{i-1} \cdots Z_1\big)\big(Z_1 \cdots Z_{j-1} c_j\big) \overset{Z_i^2=1,\, i<j}{=} c_i^\dagger\, Z_i \cdots Z_{j-1}\, c_j \\ &= Z_{i+1} \cdots Z_{j-1}\, c_i^\dagger Z_i c_j \overset{\{Z_i, c_i^\dagger\}=0}{=} -Z_i \cdots Z_{j-1}\, c_i^\dagger c_j = +Z_i \cdots Z_{j-1}\, c_j c_i^\dagger, \end{aligned} \tag{81}$$

and

$$\begin{aligned} a_j a_i^\dagger &= \big(Z_1 \cdots Z_{j-1} c_j\big)\big(c_i^\dagger Z_{i-1} \cdots Z_1\big) \\ &= Z_i \cdots Z_{j-1}\, c_j c_i^\dagger\, Z_{i-1}^2 \cdots Z_1^2 \qquad ([Z_j, c_\ell^{(+)}] = 0,\, j \neq \ell) \\ &= Z_i \cdots Z_{j-1}\, c_j c_i^\dagger. \end{aligned} \tag{82}$$

Both reduce to the same expression, so $a_j a_i^\dagger - a_i^\dagger a_j = 0$ for $i < j$, i.e. $[a_j, a_i^\dagger] = 0$; the case $i > j$ follows straightforwardly. □

Claim: $[a_i, a_i^\dagger] = 1 - 2n_i$ **with** $n_i = c_i^\dagger c_i$**.**

Proof. Using eq. (65) and $Z_k^2 = \mathbb{1}$,

$$a_i^\dagger a_i = c_i^\dagger \, Z_{i-1} \cdots Z_1 Z_1 \cdots Z_{i-1} \, c_i = c_i^\dagger c_i = n_i, \tag{83}$$

$$a_i a_i^\dagger = Z_1 \cdots Z_{i-1} \, c_i c_i^\dagger \, Z_{i-1} \cdots Z_1 = c_i c_i^\dagger = 1 - c_i^\dagger c_i, \tag{84}$$

so that

$$[\hat{a}_i, \hat{a}_i^\dagger] = a_i a_i^\dagger - a_i^\dagger a_i = (1 - c_i^\dagger c_i) - c_i^\dagger c_i = 1 - 2\hat{n}_i. \tag{85}$$

□

5 Minimizing complex functions with Wirtinger derivatives

5.1 Definition

Define for $z = x + iy \in \mathbb{C}$ with $x, y \in \mathbb{R}$

$$\frac{\partial}{\partial z} \equiv \frac{1}{2}\Big(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\Big) \qquad \& \qquad \frac{\partial}{\partial \bar{z}} = \frac{1}{2}\Big(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\Big) \tag{86}$$

for functions of one complex variable, and

$$\frac{\partial}{\partial z_j} = \frac{1}{2}\Big(\frac{\partial}{\partial x_j} - i\frac{\partial}{\partial y_j}\Big) \qquad \& \qquad \frac{\partial}{\partial \bar{z}_j} = \frac{1}{2}\Big(\frac{\partial}{\partial x_j} + i\frac{\partial}{\partial y_j}\Big) \tag{87}$$

for functions of multiple complex variables.

5.2 Agreement with the complex derivative

If f is complex differentiable at a point (i.e. the limit $f'(z_0) = \lim_{z\to z_0} \frac{f(z)-f(z_0)}{z-z_0}$ exists), then $\frac{\partial f}{\partial z}$ agrees with the complex derivative $\frac{df}{dz}$.

Proof. For $f(z) = u(z) + iv(z)$ complex differentiable,

$$\begin{aligned}\frac{\partial f}{\partial z} &\overset{(86)}{=} \frac{1}{2}\Big(\frac{\partial f}{\partial x} - i\frac{\partial f}{\partial y}\Big) = \frac{1}{2}\Big(\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} - i\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}\Big) \\ &\overset{(86)}{=} \frac{\partial u}{\partial z} + i\frac{\partial v}{\partial z} \overset{\text{Def.}}{=} \frac{df}{dz}.\end{aligned} \tag{88}$$

$\square$

5.3 The $\bar{z}$-derivative and Cauchy–Riemann

Furthermore,

$$\begin{aligned}0 \overset{!}{=} \frac{\partial f}{\partial \bar{z}} = \frac{1}{2}\Big(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}\Big) &= \frac{1}{2}\Big(\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} + i\frac{\partial u}{\partial y} - \frac{\partial v}{\partial y}\Big) \\ \iff \quad \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad &\wedge \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y},\end{aligned} \tag{89}$$

so that $0 = \dfrac{\partial f}{\partial \bar{z}}$ is equivalent to the Cauchy–Riemann equations, i.e. the necessary & sufficient condition for f to be complex differentiable.

5.4 Minimizing a real-valued function

Minimize a real-valued function[1]

$$f\colon \mathbb{C} \to \mathbb{R} \tag{90}$$

$$f(z) = f(x, y), \qquad (z = x + iy) \tag{91}$$

via the condition

$$\nabla f(x, y) = \begin{pmatrix} \partial_x f \\ \partial_y f \end{pmatrix} \stackrel{!}{=} 0. \tag{92}$$

Then

$$\partial_x f \stackrel{!}{=} 0 \stackrel{!}{=} \partial_y f \quad \Longleftrightarrow \quad \frac{\partial f}{\partial \bar{z}} = \frac{1}{2}\Big(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}\Big) \stackrel{!}{=} 0, \tag{93}$$

so one can minimize a real-valued function by imposing $\boxed{\dfrac{\partial f}{\partial \bar{z}} = 0}$.

5.5 References

- *Wirtinger derivatives*, English Wikipedia.

[1] Minimizing a complex-valued f does not make sense because $\mathbb{C}$ has no order.

6 The density matrix, entanglement entropy and their connection to singular value decomposition (SVD)

6.1 Main goal

Prove that for λ_i the eigenvalues of the reduced density matrix and s_i the Schmidt values,

$$\boxed{S_{\text{Neum}} \equiv \mathrm{Tr}\big(\rho_A \log \rho_A\big) = \sum_i \lambda_i \log \lambda_i = \sum_i s_i^2 \log s_i^2}. \tag{94}$$

6.2 The density operator

Let $\{p_i, |\psi_i\rangle\}$ be an ensemble of pure states, describing a system that is in one of the states $|\psi_i\rangle$ with probability p_i, i.e. the system is not fully known. Define the density operator as

$$\rho = \sum_i p_i \, |\psi_i\rangle\langle\psi_i| \,, \qquad p_i \in \mathbb{R} \;\; \& \;\; \sum_i p_i = 1, \tag{95}$$

which is a positive semi-definite operator of trace one (see for example Nielsen & Chuang).

6.3 Observables on a composite system

Take a system A with some observable M, and a system B, giving the composite system AB (see Box 2.6). Assume we can measure M on system A; then a corresponding measurement on AB is $\tilde{M} = M \otimes \mathbb{1}_B$.

Proof. Take AB to be prepared in $|m\rangle\,|\psi\rangle$ with $M\,|m\rangle = m\,|m\rangle$. Applying the measurement to $|m\rangle\,|\psi\rangle$ should yield m with probability one, i.e. the corresponding projector for $\tilde{M}$ is $P_m \otimes \mathbb{1}_B$. By the spectral theorem, $\tilde{M} = \sum_m m\, P_m \otimes \mathbb{1}_B = M \otimes \mathbb{1}_B$ for a general state of AB. □

6.4 The reduced density matrix as a partial trace

Measurement averages must be the same whether computed via the full ρ^{AB} or via some ρ^A associated to system A, i.e.

$$\mathrm{tr}\big(M\rho^A\big) \stackrel{!}{=} \mathrm{tr}\big(\tilde{M}\rho^{AB}\big) = \mathrm{tr}\big((M \otimes \mathbb{1}_B)\rho^{AB}\big). \tag{96}$$

Take $\rho^A = \mathrm{tr}_B(\rho^{AB})$ where

$$\mathrm{tr}_B\big(|a_1\rangle\langle a_2| \otimes |b_1\rangle\langle b_2|\big) \equiv |a_1\rangle\langle a_2| \; \mathrm{tr}\big(|b_1\rangle\langle b_2|\big) \qquad \& \qquad \mathrm{tr}_B(\cdot) \text{ linear.} \tag{97}$$

Write ρ^{AB} by using the eigenbasis of M for system A with $M\,|i\rangle = m_i\,|i\rangle$ and $\langle i|j\rangle = \delta_{ij}$:

$$\rho^{AB} = \sum_\alpha p^\alpha \, |\psi\rangle\langle\psi|^\alpha = \sum_\alpha p^\alpha \sum_{ijk\ell} c_{ij}^\alpha (c_{k\ell}^\alpha)^* \; |i\rangle_A \, |j\rangle_B \, \langle\ell|_B \, \langle k|_A \,, \tag{98}$$

$$\rho^A = \sum_\alpha \sum_{ijk\ell} p^\alpha c_{ij}^\alpha (c_{k\ell}^\alpha)^* \; |i\rangle\langle k|_A \underbrace{\mathrm{tr}\big(|j\rangle\langle\ell|_B\big)}_{=\delta_{j\ell}} = \sum_{\alpha,ijk} p^\alpha c_{ij}^\alpha (c_{kj}^\alpha)^* \; |i\rangle\langle k|_A \,. \tag{99}$$

Therefore

$$\operatorname{tr}\big(M\rho^A\big) = \sum_{\alpha,ijk} p^\alpha c^\alpha_{ij} (c^\alpha_{kj})^* \underbrace{\operatorname{tr}\big(m_i \left|i\right\rangle\!\left\langle k\right|_A\big)}_{=m_i\delta_{ik}} = \sum_{\alpha,ij} p^\alpha m_i\, c^\alpha_{ij} (c^\alpha_{ij})^*, \tag{100}$$

and similarly $\operatorname{tr}\big((M\otimes \mathbb{1}_B)\rho^{AB}\big) = \sum_{\alpha,ij} p^\alpha m_i\, c^\alpha_{ij}(c^\alpha_{ij})^*$, which proves the claim. One can also easily show that this is the only map that conserves consistency (see Box 2.6).

6.5 Reduced density matrix of a pure state

Now examine the reduced density matrix for a pure state,

$$\rho_{AB} = \left|\psi\right\rangle\!\left\langle\psi\right| = \sum_{ijk\ell} c_{ij}(c_{k\ell})^* \left|i\right\rangle_A \left|j\right\rangle_B \left\langle\ell\right|_B \left\langle k\right|_A , \tag{101}$$

$$\rho_A = \operatorname{tr}_B(\rho_{AB}) = \sum_{ijk} c_{ij}(c_{kj})^* \left|i\right\rangle\!\left\langle k\right|_A , \tag{102}$$

such that the matrix representation is

$$\boxed{\rho_A = \psi\psi^\dagger}, \tag{103}$$

using that the c_{ij} parametrize the matrix $\psi \in \mathbb{C}^{d_A\times d_B}$, i.e. $\left|\psi\right\rangle = \sum_{ij} c_{ij} \left|i\right\rangle_A \left|j\right\rangle_B$.

6.6 Connection to the SVD

Now use the SVD on ψ:

$$\psi = USV^\dagger \qquad \text{with } U\in\mathbb{C}^{d_A\times d_A},\ V\in\mathbb{C}^{d_B\times d_B} \text{ unitary.} \tag{104}$$

Then, using $V^\dagger V = \mathbb{1}$,

$$\rho_A = USV^\dagger VSU^\dagger = US^2U^\dagger, \qquad S^2 = \operatorname{diag}(s_1^2, s_2^2, \dots) \in \mathbb{R}^{d_A\times d_A}_{\geq 0}. \tag{105}$$

Furthermore, since ρ_A is hermitian we can diagonalize it with $\rho_A = ODO^\dagger = US^2U^\dagger$ and $D = \operatorname{diag}(\lambda_1,\dots)$, so that

$$\boxed{\lambda_i = s_i^2}, \tag{106}$$

i.e. the eigenvalues are given by the (squared) Schmidt values.

6.7 Functions of the density matrix via the spectral theorem

Since ρ_A is hermitian, we can use the spectral theorem to write it as $\rho_A = \sum_i \lambda_i P_i$ where λ_i are its eigenvalues and the eigenvectors $\left|u_i\right\rangle$ give $P_i = \left|u_i\right\rangle\!\left\langle u_i\right|$, since

$$\rho_A = UDU^\dagger = \big(\left|u_1\right\rangle, \dots, \left|u_n\right\rangle\big) \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} \begin{pmatrix} \left\langle u_1\right| \\ \vdots \\ \left\langle u_n\right| \end{pmatrix}, \qquad \left\langle u_i | u_j\right\rangle = \delta_{ij}. \tag{107}$$

Thus a function $f(\cdot)$ applied to ρ_A gives

$$\begin{aligned} f(\rho_A) &= \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \rho_A^n \overset{(*)}{=} \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \sum_i \lambda_i^n \, |u_i\rangle\langle u_i| \\ &= \sum_i \Big(\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \lambda_i^n \Big) |u_i\rangle\langle u_i| = \sum_i f(\lambda_i) \, |u_i\rangle\langle u_i| = U \operatorname{diag}\big(f(\lambda_i)\big) U^\dagger, \end{aligned} \tag{108}$$

where $(*)$ uses $\rho_A^n = \sum_i \lambda_i^n \, |u_i\rangle\langle u_i|$ (e.g. $\rho_A \rho_A = \sum_i \lambda_i^2 \, |u_i\rangle\langle u_i|$).

6.8 The Von Neumann entanglement entropy

Therefore, for the Von Neumann entanglement entropy,

$$\begin{aligned} S = \operatorname{Tr}\big(\rho_A \log \rho_A\big) &= \operatorname{Tr}\big(\rho_A \, U \log(\operatorname{diag}(\lambda_i)) \, U^\dagger\big) \\ &= \operatorname{Tr}\big(U \operatorname{diag}(\lambda_i) \underbrace{U^\dagger U}_{=\mathbb{1}} \log(\operatorname{diag}(\lambda_i)) U^\dagger\big) \\ &= \sum_i \lambda_i \log \lambda_i = \boxed{\sum_i s_i^2 \log s_i^2}. \end{aligned} \tag{109}$$

S is zero for two unentangled systems and non-zero for entangled systems.

6.9 Example

In our case, for example, $L = 2$, $N_f = 1$, $N_{\mathrm{ph}} = 1$:

$$\{|01\rangle, |10\rangle\} \;\&\; \{|0\rangle, |1\rangle\} \quad \text{as bases of } \mathcal{H}_f \;\&\; \mathcal{H}_{\mathrm{ph}}, \tag{110}$$

$$\Rightarrow \{|01, 0\rangle \, ; \; |01, 1\rangle \, ; \; |10, 0\rangle \, ; \; |10, 1\rangle\} \quad \text{as basis of } \mathcal{H}_f \otimes \mathcal{H}_{\mathrm{ph}}, \tag{111}$$

$$\Rightarrow |\psi\rangle = \sum_{i=1}^{2} \sum_{j=1}^{2} \psi_{ij} \, |i\rangle_f \, |j\rangle \quad \text{with } (\psi)_{ij} = \psi_{ij} \;\&\; \psi \in \mathbb{C}^{2\times 2}, \tag{112}$$

$$\Rightarrow S(\rho_A) = \sum_\ell s_\ell^2 \log s_\ell^2. \tag{113}$$

Note that since $\operatorname{tr} \rho^{AB} = \operatorname{tr} \rho^A = \operatorname{tr} \rho^B = 1$ we must have $\sum_\ell \lambda_\ell = \sum_\ell s_\ell^2 = 1$.

6.10 References

- M. Nielsen & I. Chuang, *Quantum Computation and Quantum Information* (density operator, partial trace; Box 2.6).
- U. Schollwöck, *The density-matrix renormalization group in the age of matrix product states*, beginning of Chapter 4 (MPS).

7 Quick refresher on spin

Spin can be described with the operators $\hat{S}_x, \hat{S}_y, \hat{S}_z$ that obey the following commutation relations:

$$\boxed{[\hat{S}_j, \hat{S}_k] = i\hbar\, \epsilon_{jk\ell} \hat{S}_\ell} \qquad (\hbar = 1), \tag{114}$$

with $i, j, k = x, y, z = 1, 2, 3$ and using the Einstein summation convention. One can define raising and lowering operators as

$$\boxed{\hat{S}_\pm = \hat{S}_x \pm i\hat{S}_y} \qquad \text{with} \qquad \boxed{[S_-, S_+] = -2S_z}. \tag{115}$$

Proof.

$$\begin{aligned}
[S_-, S_+] &= [S_x - iS_y,\ S_x + iS_y] \\
&= [S_x, S_x] + i[S_x, S_y] - i[S_y, S_x] + [S_y, S_y] \\
&= 2i[S_x, S_y] \overset{(114)}{=} 2i \cdot i\, \epsilon_{12\ell} S_\ell = -2\, \epsilon_{12\ell} S_\ell = -2S_z,
\end{aligned} \tag{116}$$

where the first step used the bilinearity of the commutator and the last step uses $\epsilon_{jk\ell} = 0$ if two indices are equal, leaving $\epsilon_{123} S_3 = S_z$. □

For spin-$\frac{1}{2}$ particles, one can write

$$S_j = \frac{\hbar}{2} \sigma_j \qquad \text{with } \sigma_j \text{ the Pauli matrices, } j = x, y, z, \tag{117}$$

and often again $\hbar = 1$.

8 Quick refresher on unitary transformations

Let's say we have some Hamiltonian $\hat{H}$, which we examine in a basis $\{|v_j\rangle\}$ with $j = 1, \ldots, \dim(\mathcal{H})$ and $\hat{H} \colon \mathcal{H} \to \mathcal{H}$. The matrix representation of the operator $\hat{H}$ in this basis is thus given by

$$H_{ij} = \left\langle v_i \middle| \hat{H}\, v_j \right\rangle = \langle v_i | \, \hat{H} \, | v_j \rangle \, . \tag{118}$$

Applying a unitary transformation U (with $U^\dagger U = \mathbb{1} = UU^\dagger$), i.e.

$$|v_j\rangle \to |\tilde{v}_j\rangle = U \, |v_j\rangle \qquad \& \qquad \hat{H} \to \tilde{\hat{H}} = U \hat{H} U^\dagger, \tag{119}$$

yields the matrix elements

$$\tilde{H}_{ij} = \langle \tilde{v}_i | \, \tilde{\hat{H}} \, |\tilde{v}_j\rangle = \langle v_i | \, \underbrace{U^\dagger U}_{=\mathbb{1}} \, \hat{H} \, \underbrace{U^\dagger U}_{=\mathbb{1}} \, |v_j\rangle = \langle v_i | \, \hat{H} \, |v_j\rangle = H_{ij}. \tag{120}$$

Hence the matrix representation of an operator does *not* change under a unitary transformation.

9 Quick refresher on the current in electrodynamics

9.1 Maxwell's equations

Given Maxwell's equations,

$$\vec{\nabla} \cdot \vec{E}(\vec{r}, t) = \frac{1}{\varepsilon_0}\, \rho(\vec{r}, t) \tag{121}$$

$$\vec{\nabla} \cdot \vec{B}(\vec{r}, t) = 0 \tag{122}$$

$$\vec{\nabla} \times \vec{E}(\vec{r}, t) = -\frac{\partial}{\partial t} \vec{B}(\vec{r}, t) \tag{123}$$

$$\vec{\nabla} \times \vec{B}(\vec{r}, t) = \frac{1}{c^2}\frac{\partial}{\partial t} \vec{E}(\vec{r}, t) + \frac{1}{\varepsilon_0 c^2}\, \vec{j}(\vec{r}, t) \tag{124}$$

9.2 Continuity equation

From $\varepsilon_0\, \partial_t\, (eq.\ (121))$, using $\partial_x \partial_t = \partial_t \partial_x$,

$$\frac{\partial}{\partial t} \rho(\vec{r}, t) = \varepsilon_0\, \vec{\nabla} \cdot \big(\partial_t \vec{E}(\vec{r}, t)\big), \tag{125}$$

and from $\varepsilon_0 c^2\, \vec{\nabla} \cdot (eq.\ (124))$,

$$\begin{aligned} \vec{\nabla} \cdot \vec{j}(\vec{r}, t) &= \vec{\nabla} \cdot \big(\vec{\nabla} \times \vec{B}(\vec{r}, t)\big) - \varepsilon_0\, \vec{\nabla} \cdot \big(\partial_t \vec{E}(\vec{r}, t)\big) \\ &= -\varepsilon_0\, \vec{\nabla} \cdot \big(\partial_t \vec{E}(\vec{r}, t)\big) \qquad (\mathrm{div}(\mathrm{rot}\, \vec{F}) = 0). \end{aligned} \tag{126}$$

Adding the two gives the continuity equation

$$\frac{\partial}{\partial t} \rho(\vec{r}, t) + \vec{\nabla} \cdot \vec{j}(\vec{r}, t) = 0. \tag{127}$$

9.3 Point-particle solutions

Solutions of the continuity equation:

$$\rho(\vec{r}, t) = \sum_\alpha q_\alpha\, \delta\big(\vec{r} - \vec{r}_\alpha(t)\big) \tag{128}$$

$$\vec{j}(\vec{r}, t) = \sum_\alpha q_\alpha\, \vec{v}_\alpha(t)\, \delta\big(\vec{r} - \vec{r}_\alpha(t)\big) \tag{129}$$

Proof. Note that

$$\begin{aligned} \partial_t\, \delta\big(\vec{r} - \vec{r}_\alpha(t)\big) &= \partial_t \Big(\delta\big(r_x - r_x^\alpha(t)\big)\delta\big(r_y - r_y^\alpha(t)\big)\delta\big(r_z - r_z^\alpha(t)\big)\Big) \\ &= -\sum_{i=x,y,z} \dot{r}_i^\alpha(t)\, \partial_{r_i} \delta\big(r_i - r_i^\alpha(t)\big) = -\dot{\vec{r}}_\alpha(t) \cdot \vec{\nabla}\, \delta\big(\vec{r} - \vec{r}_\alpha(t)\big), \end{aligned} \tag{130}$$

where we used the chain rule in the second line and

$$\vec{\nabla} \cdot \Big(\vec{v}_\alpha(t)\, \delta\big(\vec{r} - \vec{r}_\alpha(t)\big)\Big) = \sum_{i=x,y,z} \partial_{r_i} \Big(v_i^\alpha(t)\, \delta\big(\vec{r} - \vec{r}_\alpha(t)\big)\Big)$$
$$= \sum_{i=x,y,z} v_i^\alpha(t)\, \partial_{r_i} \delta\big(\vec{r} - \vec{r}_\alpha(t)\big) = \vec{v}_\alpha(t) \cdot \vec{\nabla}\, \delta\big(\vec{r} - \vec{r}_\alpha(t)\big). \qquad (131)$$

Therefore

$$\frac{\partial}{\partial t}\rho(\vec{r}, t) = -\sum_\alpha q_\alpha\, \dot{\vec{r}}_\alpha(t) \cdot \Big(\vec{\nabla}\, \delta\big(\vec{r} - \vec{r}_\alpha(t)\big)\Big) = -\vec{\nabla} \cdot \vec{j}(\vec{r}, t), \qquad (132)$$

which is (127). □

Note that these solutions are not unique (one equation for four components).